Has US Population Growth Stopped Following Income?

Income, migration and the stalling of big cities, 1880–2020
(progress report)

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DATA320

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Abstract

For most of US history, population grew fastest in the states where incomes were highest. This project asks whether that is still true and, if not, when it changed. With yearly state population and income data from 1929 to 2024 (and decade data back to 1880 for context), I estimate how much faster richer states grew than poorer ones, and I test whether that relationship was the same before 1970 and after 1990. It was not: before 1970 a state with 10% higher income grew about 1.7 percentage points faster per decade, while from 1990 on income no longer predicted growth. The difference is statistically significant ($\chi^2(1) = 19.0$), and it stays significant after accounting for the fact that the cut years were chosen by searching the same data ($p \approx 0.001$). Exactly when the change happened is not settled. In decade data from 1950, nearly all of the decline is in estimated net migration rather than in births and deaths. A second analysis of 81 large metropolitan areas shows that when the old big cities fell well behind US growth in 1950–70, their metro areas mostly kept growing. Only from the 1970s did whole metro areas fall behind, almost all of them in the Northeast and Midwest. The next stage of the project will test whether housing costs and land-use rules explain these patterns.

1 Questions

Housing has become expensive in the US cities with the strongest economies. One explanation is that rules limiting new housing keep people from moving to those cities. Before testing that explanation, the project first establishes the basic facts it has to explain. This report covers two of them:

Q1. Does population still grow faster in richer states, and when did that change? If housing limits are keeping people out of high-income places, the historical tendency of population to grow toward higher incomes should have weakened.

Q2. When big cities stopped growing after 1950, did their whole region stop growing, or did only the city proper fall behind while its suburbs kept growing? City population counts alone can make a growing region look like it is in decline.

2 Data

All data are public. Table 1 lists the sources. The data pipeline is written in Python; it downloads or reads each source, converts it into one common long format, checks it and builds the analysis panels.

Table 1: Data sources.

Data	Source	Coverage
State population	US Census via IPUMS NHGIS [6]	1880–2020, every 10 years
State income per person	Klein [5]; Easterlin [2]; BEA	1880–1920; BEA 1929 on
State population and income, every year	Bureau of Economic Analysis (BEA)	1929–2024
County net migration	UW–Madison Applied Population Lab [8]; ICPSR	1950s–2010s, by decade
City population	Stanford CESTA [7]; IPUMS NHGIS	1900–2020
City land area	Census Working Paper 27 [4]; Census Gazetteer files	1950–2020
Metro areas (counties in each)	OMB 2023 metro definitions	held fixed for all years

There are 50 states plus DC. A *metro* is a 2023 metropolitan area, built from its counties, so its boundary stays the same in every year. The city analysis uses the **81 metros that contained one of the 100 largest US cities in 1950**; "the city" is that metro's largest 1950 city.

3 Methods

3.1 How strongly does growth follow income?

For each period, I regress population growth on income:

$$\underbrace{\ln P_{i,t+1} - \ln P_{i,t}}_{\text{growth of state } i} \;=\; \alpha_t \;+\; \beta \cdot \underbrace{\ln(\text{income per person})_{i,t}}_{\text{how rich state } i \text{ is}} + \varepsilon_{i,t}. \tag{1}$$

Here t is a year in the yearly data and a census year in the decade data (so $t+1$ is ten years later). The slope β measures how much faster richer states grow. All slopes in this report are scaled to growth *over ten years*: yearly slopes are multiplied by 10. For example, $\beta = 0.174$ means that a state with 10% higher income grows about $0.174 \times \ln(1.1) \approx 0.017$, or 1.7 percentage points, faster per decade. A positive β means population grows faster where incomes are higher. A β near zero means growth no longer depends on income. The term α_t is a separate intercept for each year. It absorbs anything that hit the whole country at once, such as the baby boom, wars and recessions, so β compares states only with each other in the same year. Every state counts equally, whatever its population.

3.2 The statistical test

Model. I use the yearly BEA data (1929–2024) and let the slope differ across three periods:

$$g_{i,t} = \alpha_t + \beta_{\text{before}}\, x_{i,t}\, \mathbf{1}[t < 1970] + \beta_{\text{middle}}\, x_{i,t}\, \mathbf{1}[1970 \le t < 1990] + \beta_{\text{after}}\, x_{i,t}\, \mathbf{1}[t \ge 1990] + \varepsilon_{i,t}, \tag{2}$$

where $g_{i,t}$ is growth from year t to $t+1$ and $x_{i,t}$ is log income per person. The symbol $\mathbf{1}[\cdot]$ is the *indicator function*: it equals 1 when the condition in brackets is true and 0 when it is false,

$$\mathbf{1}[t < 1970] = \begin{cases} 1 & \text{if } t < 1970, \\ 0 & \text{otherwise.} \end{cases}$$

In any given year exactly one of the three indicators equals 1, so each year's growth is explained by one slope only. For example, in 1955 the equation reduces to $g_{i,1955} = \alpha_{1955} + \beta_{\text{before}}\, x_{i,1955} + \varepsilon_{i,1955}$. There are 4,803 state-years: 51 states (including DC), growth years 1929–2023. The 1970 and 1990 cut points come from earlier break-date searches in this project on these same data. Because the cuts were chosen by looking at the data, the p-value of this test is *nominal*: it overstates the evidence. A second test, described below, accounts for the search.

Hypotheses.

$$H_0 : \beta_{\text{before}} = \beta_{\text{after}} \qquad \text{vs.} \qquad H_1 : \beta_{\text{before}} \neq \beta_{\text{after}}.$$

Test statistic. A Wald test:

$$W = \frac{(\hat{\beta}_{\text{before}} - \hat{\beta}_{\text{after}})^2}{\widehat{\text{Var}}(\hat{\beta}_{\text{before}} - \hat{\beta}_{\text{after}})} \;\sim\; \chi^2(1) \text{ under } H_0.$$

The variance is *clustered by state*. The same state appears up to 95 times, and its yearly errors are not independent: a state that grows quickly one year tends to grow quickly the next. Treating the 4,803 rows as independent would make the test too confident. Clustering treats each state's whole history as one independent unit, so there are effectively 51 observations, and it also allows the error variance to differ across states. It does not allow for shocks shared by neighbouring states. The significance level is $\alpha = 0.05$.

Accounting for the choice of cut years. The *sup-Wald* test (Andrews [1]) fits a model with a single break and tries every possible break year, leaving out the first and last 15% of years. It takes the largest Wald statistic over all those years and compares it with the distribution of that maximum when there is no break. Because it pays for having searched, its p-value is valid even though the break year was chosen from the data.

3.3 Where the change comes from

Over a decade, a state's growth is the sum of net migration and natural increase (births minus deaths). Net migration comes from published county estimates, summed to states. These are *residual* estimates: each county's population change minus its recorded births and deaths. They include immigrants from abroad. The rest of growth, total growth minus net migration, is mostly natural increase. From the 1970s on it also absorbs any change in how completely the census counted people; for the 1950s–60s that change sits in net migration instead. Because regression is linear in the outcome, β for total growth splits exactly into β for net migration plus β for the rest. To make the parts add up exactly, these decade slopes use ten-year percent change rather than log growth. Percent change gives somewhat larger slopes than the log growth in equation (1): 0.35 against 0.28 in the 1950s.

3.4 Cities versus their metros

For each metro and each period (1950–70, 1970–90, 1990–2020), I ask whether the city and the whole metro *kept up*, meaning they grew by at least half as much as the US did over the same period. US growth was 34% in 1950–70, 22% in 1970–90 and 33% in 1990–2020. Each metro then falls into one of four classes:

	Metro kept up	Metro fell behind
City kept up	growing	city-led
City fell behind	suburbanizing	stagnant

Suburbanizing means the city fell behind while its metro kept up. The labels describe population counts; they do not show that anyone moved out of the city. *Stagnant* means the whole metro fell behind. A city is *annexing* if its land area grew more than 20% over the period (a merger with its county also counts), so new residents partly came from moving the city line; otherwise it has *fixed* boundaries. Ranges around each share come from the bootstrap: I resample the metros with replacement 999 times and report the middle 90% of the results. The 81 metros are not a random sample but all metros with a 1950 top-100 city, so a range shows how much a share depends on which metros are in the set. The decision rules (what share counts as "mostly suburbanization," and so on) were written down before any class or share was computed.

4 Results

4.1 Growth stopped following income (Q1)

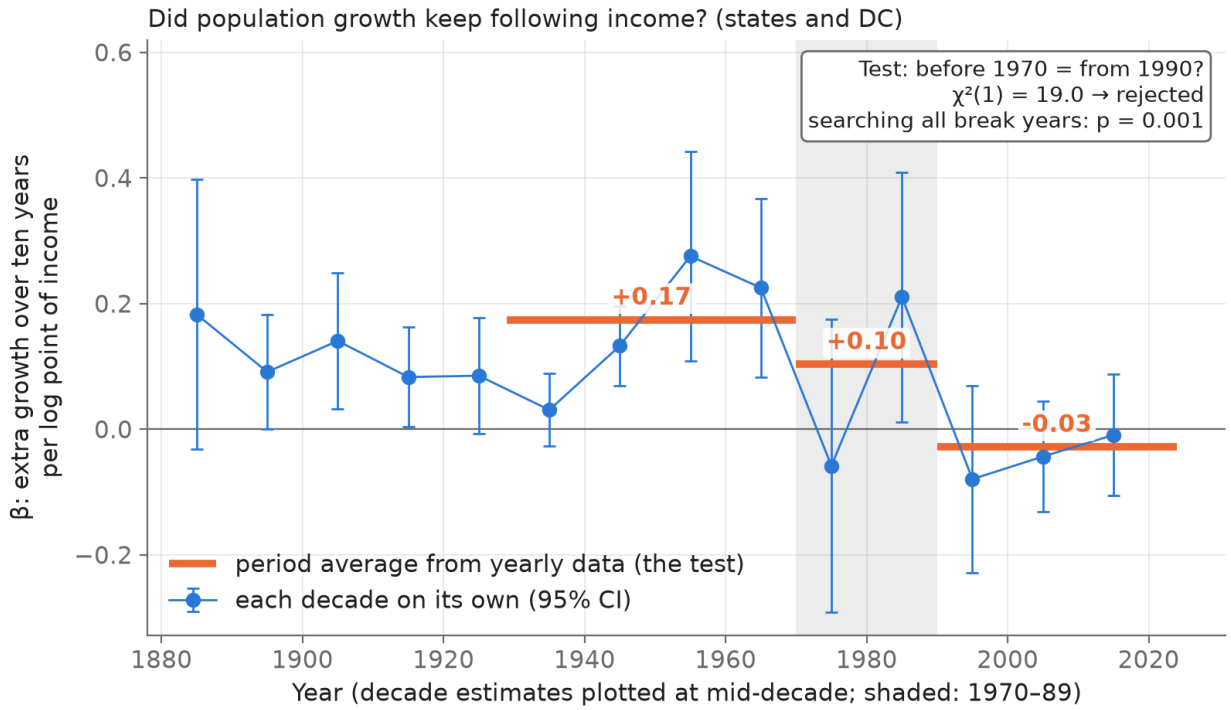


Figure 1: Population growth toward richer states, 1880–2020. Each **blue dot** is the slope β from equation (1), estimated on one decade alone (45 to 51 states; all 51 from 1950), plotted at the middle of the decade, with a 95% confidence interval. The three **orange bars** are the period slopes from the test model, equation (2), using yearly data. The grey band marks 1970–89. The box reports the test of whether the first and last orange bars are equal, and the p-value of the search over all break years.

How to read Figure 1. The vertical axis shows how much faster a richer state grew. Points above the zero line mean that population grew faster where incomes were higher; points near zero

mean that income no longer predicted growth. From 1880 to the 1960s, every decade's estimate is above zero, between 0.03 and 0.28, although several of the early intervals include zero. In the 1970s the dot falls below zero, it rises again in the 1980s, and from the 1990s on every dot is at or below zero. The orange bars summarize this: +0.17 before 1970, +0.10 in 1970–89 and −0.03 from 1990. Some decades have wide intervals because each blue dot uses only 45 to 51 data points, which is why the test pools many years.

Test result. Table 2 gives the estimates. The slope fell by 0.20 (95% confidence interval 0.11 to 0.29), with $W = 19.0$ and a nominal $p = 1.3 \times 10^{-5}$. Because the cut years were chosen from these data, that p-value is too small. The sup-Wald test, which searches over every break year, finds the best single break at 1990 with a statistic of 17.8, against a 1% critical value of 12.5. Its p-value, which accounts for the search, is 0.0008. **We reject H_0 either way**: the link between income and growth before 1970 differs from the link after 1990. In practical terms, a state with 10% higher income grew about 1.7 percentage points faster per decade before 1970. After 1990 the slope is −0.03 with a standard error of 0.05, which is not different from zero: income no longer predicted growth, but richer states did not clearly grow *slower*. This matches the direction Ganong and Shoag [3] found with different methods; they date the weakening to about 1980.

Table 2: The test: income slope by period (yearly state data, 1929–2024; slopes on the ten-year scale; standard errors clustered by state).

	Estimate	Std. error
$\hat{\beta}_{\text{before}}$ (1929–1969)	0.174	0.038
$\hat{\beta}_{\text{middle}}$ (1970–1989)	0.103	0.112
$\hat{\beta}_{\text{after}}$ (1990–2023)	−0.028	0.047
Difference, before − after	0.203	0.046
95% confidence interval	0.112 to 0.294	
Wald statistic W (χ^2, 1 d.f.)	19.0	
p-value (nominal: cut years chosen from the data)	1.3×10^{-5}	
Search over all break years: sup-Wald (best year 1990)	17.8	
critical values, 5% / 1%	8.8 / 12.5	
p-value, accounting for the search	0.0008	
Observations (state-years) / clusters (states)	4,803 / 51	

Is the result fragile? I re-ran the test seven other ways. Each version changes the assumptions in one area, and the test rejects H_0 in all of them, with the largest p-value 0.008 (Table 3). The notebook's main version for dating the change adds separate year effects for mining- and energy-producing states during the oil boom and bust; this report uses the simpler version with year effects only, and the two give the same p-value.

When did it change? The test establishes that a change happened, not when. The middle period, 1970–89, is not significantly different from either side ($p = 0.42$ against "before" and $p = 0.16$ against "after"). The decade dots show why: the 1970s look like the period after, while in the 1980s growth again followed income. The project's rule for dating the change, fixed in advance, gives no answer with the yearly data. With decade data it gives 1970, but that answer depends on a single unit: leaving out the District of Columbia turns it into a gradual change (and the yearly

Table 3: The same test under other assumptions (nominal p-values).

Version	p-value
This report's version (year effects only)	1.3×10^{-5}
Energy-producing states get their own year effects in 1973–1990 (oil boom and bust)	1.3×10^{-5}
... same, with a hand-picked list of energy states	1.2×10^{-5}
... same, oil years 1973–1986 only	1.3×10^{-5}
States compared only within their census division (9 regions)	0.008
Income standardized within each year (with energy-state year effects)	4.9×10^{-8}
World War II years (1940–46) left out (with energy-state year effects)	2.2×10^{-4}
Each state left out in turn (51 versions; largest p shown)	1.7×10^{-4}

answer into 1990), while leaving out any other state changes nothing. DC is unusual, a city with no suburbs inside its border. The timing is therefore still open.

Most of the change is in net migration. Table 4 splits the decade slopes into net migration and the rest. These slopes are not on the same scale as Table 2: they use decade data from 1950, so "before" means the 1950s–60s only, when the slope was highest, and they use percent change. Of the 0.37 drop in the total slope between the 1950s–60s and the 1990s–2010s, 0.35 is in estimated net migration. Births and deaths followed income only weakly in every period, and their slope did not change significantly ($p = 0.65$). The decade result is the same without the six main immigrant gateway states (California, New York, New Jersey, Illinois, Florida and Texas).

Table 4: Income slope of decade growth and its two parts (50 states and DC; ten-year percent change; the slopes add up exactly: total = net migration + the rest).

	1950s–60s	1970s–80s	1990s–2010s
Total growth	0.320	0.096	−0.047
Net migration (residual estimates)	0.291	0.065	−0.058
Births − deaths (+ census coverage change)	0.029	0.031	0.012

4.2 Big cities: city first, whole metros later (Q2)

How to read Figure 2. A value of 0.5 means half the metro lived in the main city. In 1940 the typical big city held about half its metro's population; by 2020 it held about a quarter. The fall is steepest between 1950 and 1980 and flattens after 2000. The share can fall either because the city lost people or because the suburbs grew faster, so this figure alone can't separate decline from suburbanization. Figure 3 does. Cities that annexed land, such as Houston and Austin, kept a higher share for longer.

How to read Figure 3. Read the colors from the bottom up. In 1950–70 (left panel), the largest group is orange, *suburbanizing*: 43 of 81 metros (53%) kept up while their main city fell behind. Green, *stagnant* (the whole metro fell behind), is only 6 metros. In 1970–90 green becomes the largest group, 35 of 81 (43%). In 1990–2020 it shrinks a little, to 28 of 80, while blue (*growing*, city and metro both kept up) recovers. The right panel, with a fixed 5%-per-decade bar, shows a similar pattern, although there suburbanizing and growing are nearly tied in 1950–70.

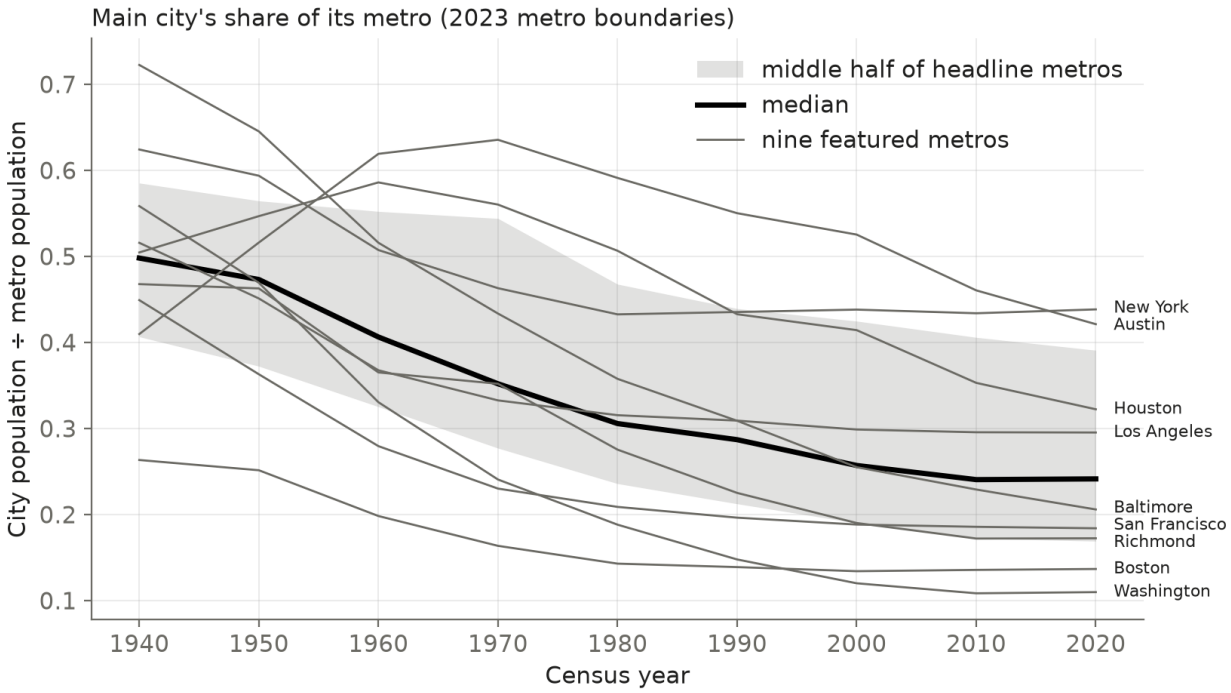


Figure 2: The main city's share of its metro's population, 1940–2020. Each thin grey line is one of nine featured metros, labelled at the right. The **black line** is the median across all 81 metros (the "headline" metros: those with a 1950 top-100 city), and the **grey band** holds the middle half of them (25th to 75th percentile). Metro boundaries are the 2023 definitions throughout, so a falling share means the city grew more slowly than the rest of its region, not that the region was redrawn.

Results, with 90% bootstrap ranges.

- **Mostly suburbanization in 1950–70, for cities that fell well behind.** Of the 49 cities that fell behind, 88% (79–96%) were in metros that kept up; 33 of the 49 lost population. The share is 78–93% under the alternative definitions fixed in advance. Stricter bars, tried afterwards, weaken it: 65% (54–75%) when "kept up" means three-quarters of US growth, and 52% (41–63%) at the full US rate. So the result holds for cities that fell well behind the US, not for every city that grew more slowly than the US.
- **Fixed-boundary cities fell behind far more often.** Among metros that kept up in 1950–70, 30 of 33 fixed-boundary cities (91%) fell behind, against 8 of 36 annexing cities (22%); 5 cities with unknown land area are left out. The difference is 0.69 (0.54–0.82). Part of this is built in, since people in annexed land count as city growth.
- **Whole metros fell behind later.** The share of metros that fell behind rose from 9% (1950–70) to 43% (1970–90), then 36% (1990–2020). The rise is 0.35 (0.25–0.44), and it holds at the stricter bars too. It is almost all in the Northeast and Midwest (12% → 77% → 61%); in the South and West it stayed at 5–8%. This pattern belongs to the old big metros: across all other US metros the share was 19%, 21% and 26%. So choosing the 1950 top 100 makes the rise larger, not smaller.

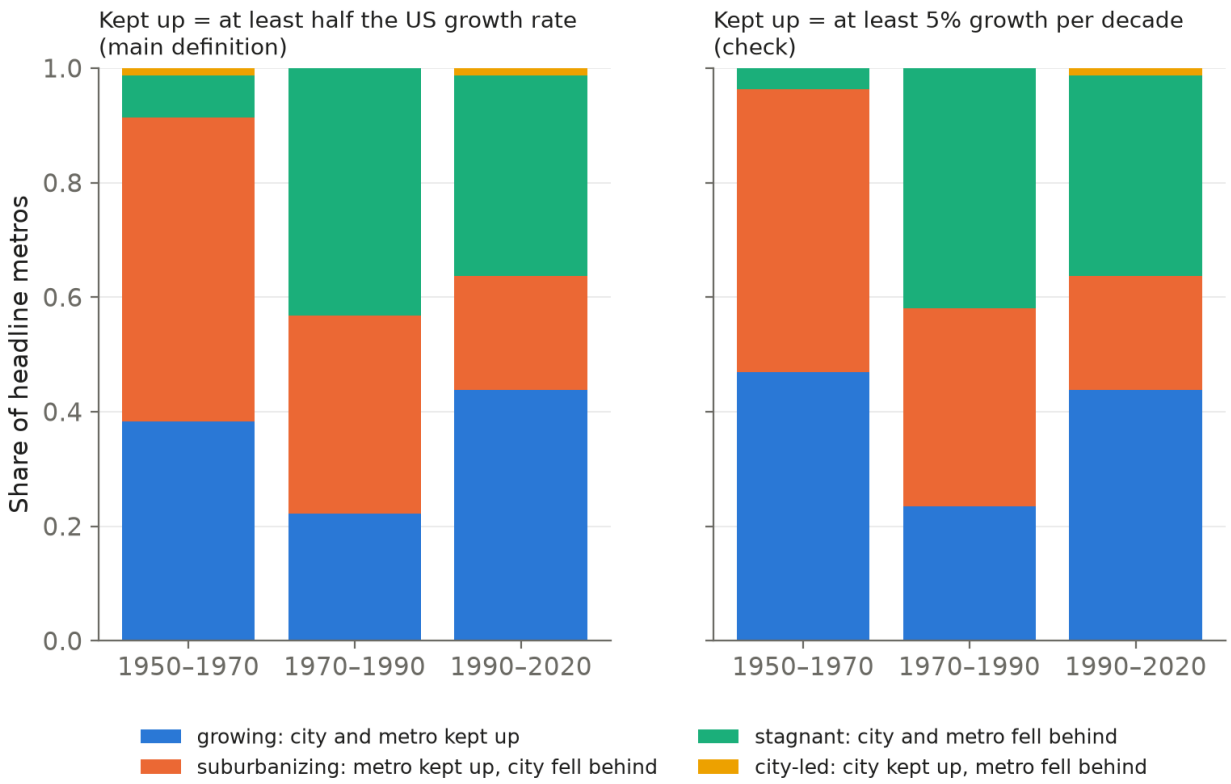


Figure 3: How the 81 big-city ("headline") metros grew, by period. Each bar divides the metros into the four classes defined in Section 3.4; the legend repeats the definitions. Bar heights are shares, so each bar sums to 1. In 1990–2020 one metro (Denver) has no class, because a county split across its boundary, so that bar covers 80 metros. **Left:** "kept up" means growth of at least half the US rate over the period (the main definition). **Right:** "kept up" means at least 5% per decade, a fixed bar that does not depend on how fast the US grew (a check).

5 Limitations

- **Association, not cause.** The test shows that the link between income and growth changed. It does not show *why*. Housing costs are one explanation; others include retirement migration to the Sun Belt, air conditioning, and the decline of manufacturing, which the regional pattern in Section 4.2 suggests: the Northeast and Midwest were hit hardest.
- **Timing.** The data agree that the change happened, but not on when. The decade answer of 1970 depends on whether DC is included.
- **States only.** The test uses states. The yearly break search on metro areas finds no significant break, but metro income data start only in 1969, too late to see a change around 1970. Decade-level break tests on states, with only 51 data points per decade, find no break in their main version either; they do find one when income is standardized within each decade.
- **Equal weights.** Every state counts equally in the regressions, whatever its size; a population-weighted version has not been run.
- **Net migration is estimated**, not counted, and includes immigration from abroad.
- **Income data before 1929** come from historical estimates (Klein; Easterlin) that are less

precise than BEA's; the test itself uses only BEA data from 1929. Income is not adjusted for differences in living costs between states.

- **Fixed 2023 metro boundaries** include counties that were rural in 1950. Using only the central counties changes the 1950–70 share by one metro (86% instead of 88%) and leaves all three answers the same.
- **Land areas** come from two sources that measure slightly differently (1950–80 and 1990–2020), which adds some noise to the annexing classification.

6 Next steps

1. Test the housing explanation directly: do metros that permit less housing (Census Building Permits Survey) or have stricter land-use rules (Wharton index) attract fewer people for the same income growth, and have their prices risen faster relative to income?
2. Compare with other countries (OECD metro data), to see whether the US is unusual.
3. A case study of a policy change: Auckland's 2016 upzoning and the construction that followed.

Use of AI tools

This report was written with the help of Claude Code, an AI coding assistant made by Anthropic. It was used to write and test the data pipeline and analysis code, to make the figures and to draft the text of this report. Three independent AI reviewers then checked the notebooks and this report against the data; their corrections, mainly to how strongly the results were worded, are included in this version.

Reproducibility

The project is a Python package with a data pipeline, two analysis notebooks (`1.1-lr-us-structural-break` for Q1, `1.0-lr-city-vs-metro` for Q2) and 249 automated tests that check both the code and the data, for example that city populations match published Census figures. Every number in this report comes from, or is counted from, tables that the notebooks write to `reports/tables/`.

References

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